

Homework 3

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February 8, 2026

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Problem 1. Give an example of $G \subset \mathrm{GL}(n, \mathbb{C})$ and $X \in \mathfrak{gl}(n, \mathbb{C})$ such that $e^X \in G$ but $X \notin \mathrm{Lie}(G)$.

Proof. Take $G = \mathrm{SL}(2, \mathbb{C})$ with Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ given by traceless 2×2 matrices over $\mathbb{C}$. Let $X = i\pi I$ for I the identity. The exponential is

$$e^X = \begin{pmatrix} e^{i\pi} & 0 \\ 0 & e^{i\pi} \end{pmatrix} = -I.$$

Note that $\det(-I) = -1^2 = 1$ since we take $n = 2$, and thus $-I \in \mathrm{SL}(2, \mathbb{C})$. The trace of X is

$$\mathrm{tr}(X) = i\pi + i\pi = 2\pi i \neq 0.$$

Thus, $X \notin \mathfrak{sl}(2, \mathbb{C}) = \mathrm{Lie}(G)$. ■

Problem 2. Let $A \in \mathrm{SL}(2, \mathbb{C})$ which has an eigenvalue in $\mathbb{C} \setminus \mathbb{R}$. Show that

$$A = C \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} C^{-1}$$

for some invertible matrix C .

Proof. Let $\lambda_1, \lambda_2 \in \mathbb{C}$ be the eigenvalues of A , which exist since $\mathbb{C}$ is algebraically closed and eigenvalues are a solution to a polynomial. Also since the eigenvalues solve the characteristic polynomial of A , we have that $p(t) = \det(A - tI)$ factors as $(t - \lambda_1)(t - \lambda_2) = t^2 - (\lambda_1 + \lambda_2)t + \lambda_1\lambda_2$. We then have $p(0) = \det A = \lambda_1\lambda_2$. However, $A \in \mathrm{SL}(2, \mathbb{C})$ implies $\lambda_1\lambda_2 = 1$, hence $\lambda_2 = \lambda_1^{-1}$.

Let $\lambda \in \mathbb{C} \setminus \mathbb{R}$ be an eigenvalue of A , which exists by hypothesis. Since $\lambda \notin \mathbb{R}$, we have $\lambda \neq \pm 1$, and thus $\lambda \neq \lambda^{-1}$ (as $\lambda = \lambda^{-1} \implies \lambda^2 = 1$). Since A has distinct eigenvalues, it admits a diagonalization, rendering it conjugate to the diagonal matrix of eigenvalues

$$D = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}.$$

Now consider the matrix

$$B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

with characteristic polynomial

$$q(t) = (\cos \theta - t)(\cos \theta - t) + \sin^2 \theta = \cos^2 \theta + t^2 - 2\cos(\theta)t + \sin^2 \theta = t^2 - 2\cos(\theta)t + 1.$$

The quadratic equation yields roots for q :

$$t = \frac{2\cos \theta \pm \sqrt{4\cos^2 \theta - 4}}{2} = \cos \theta \pm \sqrt{-\sin^2 \theta} = \cos \theta \pm i\sin \theta = e^{\pm i\theta}.$$

Since the complex exponential map $\exp : \mathbb{C} \rightarrow \mathbb{C} \setminus \{0\}$ is surjective, there exists $\theta \in \mathbb{C}$ such that $e^{i\theta} = \lambda$, hence B has eigenvalues λ and $e^{-i\theta} = (e^{i\theta})^{-1} = \lambda^{-1}$. Since these are distinct, B is also conjugate to D . Since conjugacy is an equivalence relation, we find that there exist matrices C such that $A = CBC^{-1}$. ■